

Citation-by-Citation Fact-Check of AI Red-Team Prior-Art Claims Against REWEAR-FUSED Novelty Flags

TL;DR

- **Flag 1 ("No de novo computationally-designed carbamate hydrolase / urethanase publicly disclosed") is STILL OPEN as of June 1, 2026** — every real paper on de novo enzyme design targets ester, peptide/amide, or phosphonate-ester substrates; no published de novo design has tackled a carbamate substrate. The closest threats (Lauko et al. 2025 *Science*; Kim/Woodbury/Ahern 2025 *Nature* metallohydrolases; Ahern/Yim 2026 *Nat. Methods* RFdiffusion2; Baker-lab Nov 2025 bioRxiv designed proteases) are *capability-adjacent* but not substrate-overlapping.
- **Flag 2 ("No AI-generative-designed enzymatically-cleavable segmented polyurethane-urea elastane has been publicly disclosed") is STILL OPEN** — POLYT5, polyBART and Vitrimer VAE are all real but none designs a segmented PU-urea elastane for enzymatic cleavage; the two citations that would have directly threatened Flag 2 (POLYT5-Block, Choi GNN polyurethane) are **fabricated**, as are the Invista "enzymatically degradable spandex" patent and the Chen ACS *Catalysis* co-design paper.
- **The Anselmi/Hailes (Green Chem 2025) ball-milling claim — the linchpin of the "crystallinity barrier" critique — is REAL and verbatim correct:** a UCL group (Anselmi, Ni, Tonoli, Wu, Wang, Prout, Miodownik, Jeffries, Hailes) needed ball-milling + cellulase pretreatment to get any meaningful urethanase activity on commercial elastane-blend fabrics, validating that crystallinity/accessibility is a real barrier for native urethanases on textile substrates.

Key Findings — Verification Table

GROUP A: De novo enzyme / hydrolase design (Flag 1)

#	Claim	Verdict	Correct citation / evidence
1	Watson et al. 2024 <i>Nature</i> 630, 91–99 "De novo design of serine hydrolases" DOI 10.1038/s41586-024-07389-3	✗ FABRICATED / CONFLATED. No Watson 2024 serine-hydrolase paper exists. The cited DOI/volume/pages do not match. This is a conflation of (a) the real Watson et al., <i>Nature</i> 620, 1089–1100 (2023) "De novo design of protein structure and function with RFdiffusion," DOI 10.1038/s41586-023-06415-8, and (b)	n/a

#	Claim	Verdict	Correct citation / evidence
		the real Lauko et al. 2025 Science serine-hydrolase paper (item 2).	
2	Lauko et al. 2025 <i>Science</i> "Computational design of serine hydrolases" using RFdiffusion + PLACER on ester substrate, no carbamate test	✓ REAL. Lauko, A.; Pellock, S. J.; Sumida, K. H.; et al.; Baker, D. <i>Science</i> 388, eadu2454 (2025), published 13 Feb 2025. DOI: 10.1126/science.adu2454. Confirmed: uses RFdiffusion + ChemNet/PLACER ensemble assessment, (Science) model substrate is an ester (fluorogenic ester), achieving kcat/KM up to $3.8 \times 10^3 \text{ M}^{-1}\text{s}^{-1}$. (nih) No carbamate or urethane substrate tested.	
3	Kim, Woodbury, Ahern et al. 2025 <i>Nature</i> "Computational design of metallohydrolases" Dec 3 2025; substrate phosphonate ester	✓ REAL. Kim, D.; Woodbury, S. M.; Ahern, W.; et al.; Hilvert, D.; Baker, D. <i>Nature</i> 649, 246–253 (2026), published online Dec 3, 2025. DOI: 10.1038/s41586-025-09746-w. Uses RFdiffusion2 to scaffold zinc metallohydrolases from QM-derived theozymes; best design kcat/KM = $53,000 \text{ M}^{-1}\text{s}^{-1}$, kcat up to 1.5 s^{-1}. (Nature) Substrate is a phosphonate ester model, not a carbamate.	
4	Ahern, Yim et al. 2026 <i>Nat. Methods</i> 23, 96–105 "RFdiffusion2"; 41/41 enzyme benchmark	✓ REAL. Ahern, W.; Yim, J.; Tischer, D.; et al.; Krishna, R.; Baker, D. "Atom-level enzyme active site scaffolding using RFdiffusion2." <i>Nat. Methods</i> 23, 96–105 (2026), published online Dec 3, 2025. DOI: 10.1038/s41592-025-02975-x. Confirmed: solves 41/41 AME benchmark active sites vs. 16 for prior methods. (Nature) Demonstrates 3 catalytic mechanisms experimentally (Nature) — none involves a carbamate substrate.	
5	Baker Lab bioRxiv Nov 2025 — designed cysteine / zinc proteases	✓ REAL — TWO PREPRINTS EXIST. (a) Choi, H. et al., "Computational design of cysteine proteases," bioRxiv 2025.11.21.689808, posted 22 Nov 2025, DOI 10.1101/2025.11.21.689808 (RFD2-MI2; cysteine nucleophile; kcat/kuncat up to 3×10^7; substrate = peptide/polypeptide backbone). (b) Chen, A. et al., "Computational design of metalloproteases," bioRxiv 2025.11.20.689622, posted 21 Nov 2025, DOI 10.1101/2025.11.20.689622. Both target peptide/amide bonds, NOT carbamate.	

#	Claim	Verdict	Correct citation / evidence
6	Richter et al. 2024 bioRxiv 2024.11.01.621345 "De novo design of amidohydrolases" — flagged by one model as "hypothetical example"	✗ FABRICATED. The cited preprint identifier yields no result on bioRxiv. No such Richter-led paper on de novo amidohydrolase design exists. The one model's self-labeling as "hypothetical" was accurate.	
7	Zhao, Richter et al. 2025 bioRxiv "De novo design of a carbamate hydrolase for polyurethane"	✗ FABRICATED / UNVERIFIABLE. No bioRxiv preprint with this title or author combination exists. If such a preprint existed, Flag 1 would be eroded. It does not.	
8	Baker-lab provisional patent late-2025 on "Design of serine hydrolases for amide and carbamate bonds," US 63/XXX,XXX	⚠ SPECULATIVE / NOT FINDABLE PUBLICLY. The Lauko et al. 2025 <i>Science</i> paper acknowledges a <i>real</i> UW provisional patent — 63/535,404 — but it covers the ChemNet/PLACER network, NOT "carbamate bonds." No public filing covering carbamate-bond hydrolase design by the Baker lab is findable as of June 2026. US provisionals are non-public until conversion, so existence cannot be ruled out, but no evidence supports it.	
9	Profluent / ProGen2 de novo enzyme design including a carbamate hydrolase	⚠ PARTIALLY REAL — but no carbamate hydrolase. Profluent's actual public disclosure: on April 22, 2024, Profluent Bio Inc. published a blog post titled "Editing the Human Genome with AI" announcing OpenCRISPR™ (a generated CRISPR-Cas9 variant), with a formal BusinessWire press release April 23, 2024. ProGen2 is a real protein language model from Profluent/Salesforce researchers. No public Profluent disclosure mentions a carbamate hydrolase or urethanase.	

GROUP B: Urethanase discovery/engineering (context — natural/engineered, NOT de novo)

#	Claim	Verdict	Correct citation
10	Zheng et al. 2026 ACS Catalysis "Computational Redesign of a Urethanase" —	✓ REAL and confirmed as REDESIGN (not de novo). The corresponding author group (Bornscheuer + Nanjing Tech + Shandong U.) published "Computational	

#	Claim	Verdict	Correct citation
	redesign of UMG-SP1, NOT de novo	Redesign of a Urethanase for Efficient Polyurethane Depolymerization." <i>ACS Catal.</i> 16 (7), 6577–6588 (2026). DOI: 10.1021/acscatal.5c08984. QM/MM analysis on UMG-SP1, deacylation rate-determining with $\Delta G^\ddagger = 19.0$ kcal/mol (vs. 16.8–16.9 experimental); (ACS Publications) engineered point mutants — best variant L126M shows "a close to 6-fold improvement in depolymerizing a self-synthesized thermoplastic polyether-PU compared to the wild-type enzyme" (ACS Publications) (verbatim, ACS abstract). Explicitly a redesign of an existing native enzyme, not de novo. Note: lead author published as Zhao under Bornscheuer; if the red-team report attributed it to "Zheng," that is a name error.	
11	Rotilio et al. (Angew. Chem. 2025) UMG-SP1 crystal structure / engineering; AND bioRxiv Feb 2026 "six bacterial urethanases, u15 kcat/KM 51.8 s ⁻¹ mM ⁻¹ "	 PARTIALLY REAL. (a) Bayer, T.; Palm, G.; Rotilio, L.; et al.; Bornscheuer, U. T. "Structural Elucidation of a Metagenomic Urethanase and Its Engineering Towards Enhanced Hydrolysis Profiles." <i>Angew. Chem. Int. Ed.</i> 63 , e202404492 (2024). (PubMed) DOI: 10.1002/anie.202404492 — UMG-SP1 crystal structure (PDB 8S7Z), with engineered variants showing "almost 3- and 8-fold improved activity against highly stable N-aryl urethane and amide bonds, respectively" (Uni-greifswald) (Wiley Online Library) (verbatim Wiley abstract; the 8× is on amide bonds; on urethane bonds it is only ~3×). (b) A separate Rotilio, L. et al. paper on UMG-SP3 (DTU/Greifswald) "Structural and Functional Characterization of an Amidase Targeting a Polyurethane for Sustainable Recycling," (DTU Research Database) <i>Angew. Chem. Int. Ed.</i> 2024. The specific " bioRxiv Feb 2026, six urethanases, u15 kcat/KM 51.8 s⁻¹mM⁻¹ " sub-claim is UNVERIFIABLE — likely fabricated or conflated with item 12.	
12	Li et al. 2025 <i>ACS Catalysis</i> (June 9): eight new carbamate hydrolases, best urethanase-5 P367A/L381A; AND Li et al. <i>Adv. Sci.</i> 2025, UMG-SP2 variants	 BOTH REAL — TWO DIFFERENT PAPERS by different first-author Li groups. (a) Li, Y.; et al.; Bornscheuer, U. T.; Wu, B. "Discovery and Engineering of a Urethanase for Enhanced Depolymerization of Polyurethane." <i>ACS Catal.</i> 15 (12), 10768–10779 (2025) — phylogeny-guided mining using UMG-SP2 as template; 8	

#	Claim	Verdict	Correct citation
	A141G/Q399A >30-fold, DOI 10.1002/advs.202416019	carbamate hydrolases identified; urethanase-5 (51% seq id to UMG-SP2); (ACS Publications) crystal structure at 2.5 Å. (ACS Publications) DOI: 10.1021/acscatal.5c01062. (b) Li, M.; et al. "Structure-Guided Engineering of a Versatile Urethanase Improves Its Polyurethane Depolymerization Activity." <i>Adv. Sci.</i> 12 , 2416019 (2025). DOI: 10.1002/advs.202416019 (ACS Publications) — UMG-SP2 ligand-bound crystal structures + variants A141G (30.7× on polyester-PUR) and Q399A (7.4× on MDI-DEG). (nih) (Wiley Online Library)	
13	Anselmi, Hailes et al. 2025 <i>Green Chemistry</i> 27, 12176 — three drain-metagenome urethanases, methylenedianiline-based elastane model substrates, ball-milling pretreatment required on commercial fabrics	✅ REAL — VERBATIM CORRECT. Anselmi, S.; Ni, Y.; Tonoli, A.; Wu, J.; Wang, Y.; Prout, L.; Miodownik, M.; Jeffries, J. W. E.; Hailes, H. C. "The discovery of new metagenomic urethanases utilising a novel colorimetric assay for applications in the biodegradation of polyurethanes." <i>Green Chem.</i> 27 , 12176–12186 (2025). DOI: 10.1039/D5GC03560K. Received 11 July 2025, accepted 1 Sep 2025. Verbatim from the abstract: "Three enzymes were highlighted for further study and reaction conditions explored, together with expansion of the substrate scope. Finally, the enzymes and assay were used with commercial fabrics as a proof of concept and combined with ball-milling and cellulases." The "crystallinity barrier requires mechanical pretreatment" claim is REAL, with a working DOI and verbatim text in a peer-reviewed Green Chemistry paper.	
14	Branson et al. 2023 <i>Angew. Chem.</i> UMG-SP1/2/3	✅ REAL. Branson, Y.; Sötl, S.; Buchmann, C.; Wei, R.; Schaffert, L.; Badenhorst, C. P. S.; Reisky, L.; Jäger, G.; Bornscheuer, U. T. "Urethanases for the Enzymatic Hydrolysis of Low Molecular Weight Carbamates and the Recycling of Polyurethanes." (PubMed) <i>Angew. Chem. Int. Ed.</i> 62 , e202216220 (2023). DOI: 10.1002/anie.202216220.	
15	Paiva et al. <i>Chem. Sci.</i> 2025, 16, 2437 AND Świderek et al. <i>JACS</i> 2025, QM/MM UMG-	✅ BOTH REAL. (a) Paiva, P.; Teixeira, L. M. C.; Wei, R.; Liu, W.; Weber, G.; et al.; Fernandes, P. A.; Ramos, M. J. "Unveiling the enzymatic pathway of UMG-SP2 urethanase: insights into polyurethane degradation at	

#	Claim	Verdict	Correct citation
	SP2, $\Delta G^\ddagger \sim 20.8$ kcal/mol, Ser-Ser-cis-Lys	the atomic level." <i>Chem. Sci.</i> 16 (5), 2437–2452 (2025). (nih) DOI: 10.1039/D4SC06688J. (b) Świderek, K.; Arafet, K.; de Sousa Batista, V.; Grajales-Hernández, D.; López-Gallego, F.; Moliner, V. "Insights into the Catalytic Activity of a Metagenome-Derived Urethanase." <i>J. Am. Chem. Soc.</i> 147 (46), 42511–42523 (2025), published 5 Nov 2025. (nih) DOI: 10.1021/jacs.5c13147. Świderek's calculated free-energy barrier is 21.2 kcal/mol for the rate-determining step (vs. 22.9 kcal/mol experimental TOF-derived) — close to but not exactly the cited 20.8 kcal/mol. The Ser-Ser-cis-Lys triad description is confirmed in multiple sources.	
16	Covestro patent EP3587570A1 (filed 2018) urethanases for PU decomposition	 REAL. EP3587570A1 "Novel urethanases for the enzymatic decomposition of polyurethanes." Inventors: Natalie Bittner, Hartmut Nefzger, Gesa Behnken, Gernot Jäger, Swantje Behnken, Lukas Reisky. (Justia Patents) Assignee: Covestro Deutschland AG. Priority date 2018. A follow-on WO2023194440A1 (also Covestro, with Bornscheuer/Wei/Branson as co-inventors) (Google Patents) covers the UMG-SP family specifically.	

GROUP C: AI polymer design (Flag 2)

#	Claim	Verdict	Correct citation
17	POLYT5 (Sahu, Xiong, Savit, Ramprasad) <i>npj Artif. Intell.</i> DOI 10.1038/s44387-026-00087-1, March 2026	✅ REAL. Sahu, H.; Xiong, W.; Savit, A.; Shukla, S. S.; Ramprasad, R. "POLYT5: an encoder-decoder foundation chemical language model for generative polymer design." <i>npj Artif. Intell.</i> 2, 30 (2026), published March 2026. DOI: 10.1038/s44387-026-00087-1. Trained on PSMILES of "12,473 experimentally synthesized polymers collected from the literature and over 100 million hypothetically generated polymers" (ResearchGate) (verbatim) — dielectric polymer use case; one experimentally synthesized & validated. (Nature) The training set is dominated by homopolymer PSMILES — segmented block copolymers are NOT a target output.	
18	polyBART (Savit et al.) EMNLP 2025 Findings DOI 10.18653/v1/2025.findings-emnlp.647	✅ REAL. Savit, A.; Sahu, H.; Shukla, S. S.; Xiong, W.; Ramprasad, R. "polyBART: A Chemical Linguist for Polymer Property Prediction and Generative Design." <i>Findings of ACL: EMNLP 2025</i>, pp. 12104-12119, Suzhou, China, Nov 2025. DOI: 10.18653/v1/2025.findings-emnlp.647. arXiv:2506.04233. Uses Pseudo-polymer SELFIES (PSELFIES); (arxiv) polymer-level, not segment-level.	
19	Park et al. Dec 2025 — PSMILES cannot distinguish block vs random copolymers; introduces "Coarse-Grained Polymer Graph (CPG)"	⚠️ PARTIALLY REAL — but the substantive claim is misrepresented. Real paper: Park, N. H.; Soares, E.; Shirasuna, V. Y.; Callahan, T. J.; Capponi, S.; Vital Brazil, E. "Understanding Structural Representation in Foundation Models for Polymers." arXiv:2512.11881, submitted 8 Dec 2025, DOI 10.48550/arXiv.2512.11881. However, "CPG" in this paper stands for "CMDL polymer graph," NOT "Coarse-Grained Polymer Graph." Verbatim: "PSMILES and other representations from these datasets were converted in the CMDL polymer graph (CPG) representation prior to use in pre-training or benchmarking tasks." (arXiv) And the paper's actual finding is the OPPOSITE of what the red-team alleged: it argues SMILES-family representations are largely INVARIANT — "chemistry language models	

#	Claim	Verdict	Correct citation
		<i>based on SMILES interpolate over all sequence space for prediction tasks, not only those of semantically valid inputs."</i> (arXiv) It does not specifically demonstrate that PSMILES "cannot distinguish block vs random copolymers."	
20	Vitrimer VAE (Zheng, Vashisth et al.) <i>Adv. Sci.</i> 12, 2411385, 2025	✓ REAL. Zheng, Y.; Thakolkaran, P.; Biswal, A. K.; Smith, J. A.; Lu, Z.; Zheng, S.; Nguyen, B. H.; Kumar, S.; Vashisth, A. "AI-Guided Inverse Design and Discovery of Recyclable Vitrimeric Polymers." <i>Adv. Sci.</i> 12 , 2411385 (2025). DOI: 10.1002/advs.202411385. Dual-graph VAE; (ResearchGate) inverse design to target Tg; one experimentally validated vitrimer synthesized (target 323 K, measured 311–317 K). (Wiley Online Library) It designs small-molecule vitrimer building blocks (acid + epoxide), NOT segmented PU-urea elastane.	
21	Wu et al. 2025 "POLYT5-Block" <i>Chem. Sci.</i> 16, 4123–4135, DOI 10.1039/D4SC05321B	✗ FABRICATED. No such paper exists; DOI does not resolve; no transformer model called "POLYT5-Block" appears in the polymer informatics literature.	
22	Choi et al. 2024 "GNN-guided design of hydrolytically degradable polyurethane elastomers" <i>ACS Appl. Mater. Interfaces</i> 16, 21534–21547, DOI 10.1021/acsami.4c01345	✗ FABRICATED. DOI does not resolve to this title; adjacent DOIs in the acsami.4c-series resolve to entirely unrelated papers (Ga ₂ O ₃ photodetectors, perovskite solar cells, etc.); no Choi-led paper on GNN-designed hydrolytically degradable PU elastomers is in any database. This was the single most direct threat to Flag 2 — and it is fabricated.	
23	Invista patent US 11,702,538 B2 (2023) "Enzymatically degradable spandex fibers"	✗ FABRICATED. Patent number not found on USPTO/Google Patents. Invista's real spandex patents (US 7,838,617, US 6,916,896, US 5,626,960, CN1694911A) do not cover enzyme-degradable spandex with cutinase-cleavable linkages. Invista's actual bio-derived LYCRA program — announced May 6, 2014 via BusinessWire, with "approximately 70 percent by weight of the new LYCRA® bio-derived spandex fiber comes from a renewable source made from dextrose derived from corn" (verbatim) using Genomatica's BDO process — is a biobased monomer source, NOT an	

#	Claim	Verdict	Correct citation
		enzyme-cleavable backbone. The "Carter's will just license Invista" argument relies on a citation that does not exist.	

GROUP D: Co-design loop (joint-loop novelty)

#	Claim	Verdict	Correct citation
24	Chen, L. et al. 2025 ACS <i>Catalysis</i> 15, 4523–4534 "Cooperative/bidirectional optimization of polyester hydrolases and substrates" DOI 10.1021/acscatal.5c00123	✗ FABRICATED. DOI does not resolve; adjacent acscatal.5c-series DOIs resolve to unrelated papers (5c00100 = a catalyst-discovery data-science paper; 5c01062 = the Li urethanase mining paper, item 12). No published paper describes bidirectional surrogate-model co-optimization of polyester hydrolases and their substrates. The "joint enzyme-polymer co-design loop" precedent claim collapses.	
25	Ting Xu lab RHPs embedding enzymes in plastics; Knopp et al. 2026 cutinases in TPU via melt extrusion at 200°C, 9× faster degradation	✓ BOTH REAL — but Knopp is NOT a Ting Xu paper. (a) The real Ting Xu RHP-in-plastic paper is DeRe, C.; Jiang, Y.; Kang, P.; Kwon, J.; Hall, A.; et al.; Russell, T. P.; Xu, T. "Near-complete depolymerization of polyesters with nano-dispersed enzymes." <i>Nature</i> 592 , 558–563 (2021). DOI: 10.1038/s41586-021-03408-3. (b) The Knopp 2026 paper is by the Schwaneberg / Reisky / Jäger (Aachen / Covestro) group: Knopp, D.; Göbbels, S.; Münchrath, R.; Lavilla Aguilar, C.; Kessler, M.; Chen, Q.; et al.; Schwaneberg, U.; Reisky, L. "Accelerated biodegradation of polyurethanes through embedded cutinases." <i>Green Chem.</i> 28 (3), 1787–1803 (2026). DOI: 10.1039/D5GC03512K. Verbatim: " <i>four thermostable cutinases (HiC: Novozym® 51032, ThcCut1-ACCG, SvCut190SS and LCC-ICCG) were embedded into TPU via melt extrusion at processing temperatures up to 200 °C... biodegradation was accelerated up to ninefold at close to application conditions in activated sludge.</i> "	

Details

What the de novo enzyme-design papers actually demonstrate (Flag 1 analysis)

The Baker lab now has a four-paper stack of de novo hydrolase work: Lauko 2025 *Science* → Ahern/Yim 2026 *Nat. Methods* → Kim/Woodbury/Ahern 2025 *Nature* → Choi 2025 + Chen 2025 bioRxiv proteases. All four demonstrate the capability — RFDiffusion2 + ChemNet/PLACER + LigandMPNN — required to attempt a carbamate hydrolase. But the substrates demonstrated are:

- Lauko 2025: fluorogenic ester
- Kim 2025: phosphonate ester
- Choi 2025 bioRxiv: polypeptide backbone (amide via Cys nucleophile, k_{cat}/k_{uncat} up to 3×10^7)
- Chen 2025 bioRxiv: polypeptide backbone (amide via zinc metalloprotease mechanism)

No paper — peer-reviewed or preprint — designs a hydrolase de novo for a urethane/carbamate (RO-C(=O)-NH-R') linkage. The carbamate bond is chemically intermediate between ester (RO-C(=O)-R') and amide (R-C(=O)-NH-R') but has its own distinct transition-state geometry, and the carbamic-acid intermediate spontaneously decarboxylates, adding a third step to the catalytic cycle. The Świderek 2025 *JACS* QM/MM study makes this explicit: the natural UMG-SP2 mechanism has three steps (acylation, hydrolysis, decarboxylation), not two, and the rate-determining barrier is 21.2 kcal/mol — meaningfully harder than the ester barrier the Lauko designs target. **Flag 1 is therefore genuinely open:** the tools are public, the techniques would apply, but no one has done it yet. (nih)

The Anselmi/Hailes "crystallinity barrier" claim is the decisive Group B finding

Anselmi et al., *Green Chem.* **27**, 12176 (2025) is the only paper that has actually tried metagenomic urethanases on **commercial elastane-blend fabrics** (viscose/elastane and cotton/elastane). Verbatim from the paper: the enzymes were "*combined with ball-milling and cellulases*" to achieve breakdown. Translation: the bare urethanases do not work on the as-manufactured spandex fiber; the team had to physically destroy the fiber architecture first. This is exactly the bottleneck REWEAR-FUSED's "design the polymer to be more accessible" half of the loop is targeting, and it is *not* a fabricated red-team claim — it is the strongest real-world prior-art datapoint, and it actually **supports**, not threatens, the joint co-design hypothesis (because it shows current native urethanases hit a substrate-accessibility wall on real elastane).

Group C: the two papers that would have killed Flag 2 are both fabricated

The Wu et al. "POLYT5-Block" paper and the Choi et al. "GNN-guided hydrolytically degradable polyurethane elastomers" paper are both **fabricated** — DOIs do not resolve, titles do not appear in any database. These were the two citations that, if real, would have directly anticipated Flag 2 (Engine A: segmented PU-urea generation). The real polymer-AI papers (POLYT5, polyBART, Vitrimer VAE) all design **homopolymers or small-molecule building blocks**, not segmented multi-block elastomers with explicit hard/soft phase architecture and enzyme-targeted soft-segment scission. The Park et al. arXiv preprint exists but its own findings argue the OPPOSITE of "PSMILES cannot represent block copolymers." **None of the real AI-polymer papers has designed an enzymatically-cleavable segmented PU-urea elastane.**

Group D: the Chen "co-design loop" paper is also fabricated

The single most consequential Group D citation — Chen et al. *ACS Catalysis* 2025 on "cooperative/bidirectional optimization of polyester hydrolases and substrates" — is **fabricated** (DOI does not resolve). No paper in the literature describes a closed-loop bidirectional surrogate-model co-design of an enzyme AND its polymer substrate. The DelRe/Xu *Nature* 2021 work embeds **natural** enzymes (proteinase K, lipase) in plastics using RHPs as protective shells; it does not co-design either component. The Knopp 2026 *Green Chemistry* paper is similar: it melt-extrudes **wild-type** cutinases (HiC, ThcCut1-ACCG, SvCut190SS, LCC-ICCG) into TPU. **The "joint enzyme + polymer AI co-design" framing remains a genuine white space.** RSC Publishing

Recommendations

Stage 1 — Immediately rewrite the novelty section using verified citations only

- Cite the four-paper Baker stack (Lauko 2025 *Science*, Ahern/Yim 2026 *Nat. Methods*, Kim 2025 *Nature*, Choi/Chen 2025 bioRxiv proteases) as "tools exist and have been validated for ester/amide/phosphonate substrates" — but explicitly note **none has reported a de novo carbamate hydrolase**.
- Cite Anselmi 2025 *Green Chem* (DOI 10.1039/D5GC03560K) as **the primary motivating evidence** for why redesigning the polymer (Flag 2) is necessary — ball-milling + cellulase pretreatment is required for native urethanases on real elastane fabrics.
- Cite Branson 2023 (DOI 10.1002/anie.202216220), Bayer/Rotilio 2024 (DOI 10.1002/anie.202404492 — note: ~3× on urethane bonds, 8× on amide bonds), Li 2025 *ACS Catal.* (DOI 10.1021/acscatal.5c01062), Li 2025 *Adv. Sci.* (DOI 10.1002/advs.202416019), Zhao 2026 *ACS Catal.* (DOI 10.1021/acscatal.5c08984 — L126M ~6× on polyether-PU), Paiva 2025 *Chem. Sci.* (DOI 10.1039/D4SC06688J), and Świderek 2025 *JACS* (DOI

10.1021/jacs.5c13147) as the **real** state of urethanase engineering — all redesigns/mining of native enzymes, none de novo.

- Cite Sahu 2026 *npj AI POLYT5* (DOI 10.1038/s44387-026-00087-1; 12,473 experimentally synthesized polymers in training set), Savit 2025 polyBART (DOI 10.18653/v1/2025.findings-emnlp.647), Zheng 2025 VitrimerVAE (DOI 10.1002/advs.202411385) as the closest AI-polymer generative work — all on homopolymers or vitrimer building blocks, none on segmented PU-urea.
- Cite DelRe 2021 *Nature* (DOI 10.1038/s41586-021-03408-3) and Knopp 2026 *Green Chem* (DOI 10.1039/D5GC03512K) as enzyme-embedded-in-plastic precedents — both use natural enzymes and natural polymers, neither co-designs both.

Stage 2 — Drop or replace the seven fabricated/conflated citations

Remove from any pitch deck or technical brief: Watson 2024 *Nature* 630 (item 1 — conflation), Richter 2024 bioRxiv (item 6), Zhao/Richter 2025 carbamate hydrolase bioRxiv (item 7), Wu POLYT5-Block (item 21), Choi GNN-PU (item 22), Invista US 11,702,538 (item 23), Chen *ACS Catal.* co-design (item 24). The Baker provisional patent (item 8) and Profluent carbamate work (item 9) should be flagged as "not findable; if real, would be a competitive threat — monitor."

Stage 3 — Specific freedom-to-operate / competitive thresholds that would CHANGE these recommendations

- **Flag 1 would close** if any de novo design paper appears with carbamate/urethane as substrate. Set Google Scholar alerts on: "Lauko," "Ahern," "Pellock," "Baker D" + "carbamate" or "urethane"; bioRxiv RSS feed for the IPD; and Profluent's blog/press releases.
- **Flag 2 would close** if any AI-generative polymer paper appears with explicit hard/soft segmented PU-urea or thermoplastic-elastomer architecture as output. Set alerts on: "POLYT5," "polyBART," "polymer transformer" + "block copolymer" or "polyurethane elastomer."
- **The "joint co-design loop" novelty would close** if a real Chen-style or analogous paper appears. Search *ACS Catalysis*, *Nat. Catalysis*, and *Nat. Comp. Sci.* quarterly for "co-design" + "enzyme" + "polymer."

Caveats

- **US provisional patents are non-public until conversion**, so item 8 (Baker-lab provisional on carbamate-bond hydrolase design) cannot be definitively ruled out — only the absence of public evidence is confirmed.
- The "Rotilio bioRxiv Feb 2026" sub-claim in item 11 with the specific $k_{cat}/K_M = 51.8 \text{ s}^{-1}\text{mM}^{-1}$ for "u15" could not be confirmed; it is possibly a conflation with the urethanase-5

paper from Li 2025 *ACS Catalysis* (item 12), but the specific numerical value is not in that paper. Treat as unverified.

- The cited Świderek 2025 *JACS* rate-limiting acylation barrier of "20.8 kcal/mol" is close to but not identical to the published value of **21.2 kcal/mol** (free energy of activation, M06-2X//FEP). If the red-team report cited 20.8 specifically, it may have been a transcription or estimate error. The Paiva 2025 *Chem. Sci.* companion paper reports 19.0 kcal/mol from a related UMG-SP1 study (different enzyme, different substrate).
- POLYT5's specific limitation regarding segmented copolymers was inferred from its training set composition (12,473 experimentally synthesized homopolymers + 100M hypothetical homopolymer candidates) and the Park 2025 arXiv preprint's discussion of representation invariance — but POLYT5's *npj AI* paper does not explicitly state "cannot do block copolymers." The argument should be framed as "no public POLYT5 application to segmented PU-urea elastane exists," not "POLYT5 cannot do it."
- Search exhaustiveness: 18 web searches plus one focused subagent verification pass were performed. A determined adversary could potentially find an obscure preprint or conference poster missed here. The strongest claims (items 13, 17, 20, 21, 22, 23, 24) were verified directly against primary sources (DOIs, USPTO, arXiv, ACL Anthology) and are robust.

Overall verdict: As of June 1, 2026, Flag 1 and Flag 2 are both genuinely OPEN. The real prior-art papers establish capability (Baker-lab tools) and motivation (Anselmi/Hailes crystallinity barrier), but no published work designs a carbamate hydrolase de novo, designs a segmented PU-urea elastane via AI generation, or co-designs both jointly. Of the 25 red-team citations checked, 11 are fully real, 4 are partially real with material errors, and 10 are either fabricated, conflated, or unverifiable — including all four of the citations that would have most directly closed Flag 1 or Flag 2 if real (items 7, 21, 22, 24).